Joe Bader

Principal Engineer building production AI agents, their evaluations, and the controls that keep them safe

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SUMMARY

Hands-on principal engineer and engineering leader with 20+ years building and running production systems, now focused on AI agents that finish real work safely. As founder of Tuner Logic, built a safety-sensitive AI assistant and the multi-agent platform that develops it, where agents author 100% of commits under sandboxing, scoped permissions, and human review. Previously led five engineering teams at Riskonnect and spent four years on Comcast platforms serving millions of customers. Earlier, technical lead at Intuitive Company, now EY Design Studio.

TECHNICAL

Agent Engineering: Model APIs (Cloudflare Workers AI, Anthropic, OpenAI), tool calling, structured outputs (Zod), context engineering, model routing, multi-agent orchestration, Agent Client Protocol (ACP), Claude Code, Codex
Evaluation & Controls: Task-based evals, tool-call trajectory scoring, regression guards, sandboxed execution, scoped bot identities, human approval gates, audit logging
Languages: TypeScript, JavaScript, Python, SQL, Bash, Ruby, .NET, PHP
Frameworks: React, Node.js, Hono, Drizzle, Ruby on Rails
Cloud & Infrastructure: Cloudflare (Workers, D1, R2), AWS, Azure, Heroku, Linux, Docker, Podman, Kubernetes, Terraform, Ansible, systemd, New Relic
CI/CD & Testing: GitHub Actions, Gitea Actions, Jenkins, CircleCI, Azure DevOps, Playwright, Vitest

EXPERIENCE

Founder & Principal Engineer *07/2025 - Present*
Tuner Logic (tunerlogic.com) | Remote

- Founded Tuner Logic, an AI-assisted SaaS for automotive performance tuning, where a wrong answer can damage an engine. Sole architect of the product and its delivery platform, from concept through production deployment (3,300+ agent-authored commits under my direction and review).
- Built the product's AI assistant as a server-side agent loop in TypeScript on Cloudflare Workers. The model answers tuning questions by calling 13 typed tools over the user's own calibration, log, and dyno data, with schema-validated output and a hard cap on tool rounds that forces a final answer.
- Kept the model away from decisions where it could do harm. Corrections and safety checks run in deterministic code that the model calls as tools, model-written values for fuel, ignition, and boost tables are withheld, and no edit is applied until the user reviews and accepts it. Unsafe requests are refused by rule, in ten languages.
- Made model selection evidence-based with a nightly evaluation that runs every candidate model through realistic tasks and scores the full sequence of tool calls with fixed checks instead of an LLM judge. Results change production routing only through a pull request that a person reviews.
- Built a model router that balances capability, speed, and cost for each task and uses production telemetry to sideline a model that starts returning malformed tool calls, empty responses, or errors, restoring it once it recovers.
- Built the multi-agent platform that develops the product: labeling an issue starts a supervised coding agent in its own isolated workspace, using any of six agent CLIs. Agents author 100% of commits, and any session can be watched or taken over from a dashboard or phone.
- Made long-running agent work recoverable: prompts are queued durably, dispatch is idempotent, and stuck sessions are reclaimed automatically. Several of these safeguards came from diagnosing live failures (leaked worker slots, an exhausted port pool, stalled reviews) and shipping fixes with regression tests.
- Limited what each agent can do. Coding and review agents have separate scoped identities, so the agent that writes code cannot approve it and the reviewer cannot push. Review agents run in per-session sandboxes without a source checkout and treat pull request content as untrusted input.
- Automated the path to production without giving up control. A merge queue lands only pull requests approved at their exact commit (250+ batches in its first two weeks), retries infrastructure failures, and isolates the offending change when a batch fails. A Python release broker keeps production credentials out of CI.
- Held the product to safety-engineering standards: a 100% backend test-coverage floor, ISO 26262-inspired coverage targets for domain code, production health checks every five minutes, and rehearsed restore drills.

Senior Manager, Software Engineering *12/2021 - 12/2025*
Riskonnect | Remote

- Led a major UI modernization using agentic AI to migrate a legacy Bootstrap interface, including inventorying, planning, and updating thousands of UI files.

- Scaled AI-assisted development practices across five teams, increasing pull request throughput by 40%.
- Led and mentored five cross-functional engineering teams (5–8 engineers each) on a core B2B SaaS platform, improving on-time delivery from 30% to 95% while meeting reliability targets.
- Reviewed architecture for high-impact initiatives, catching weak designs before implementation and reducing long-term platform risk.
- Implemented engineering metrics and review cadences that improved lead time for changes by 4x.

Lead Software Engineer (Contract) *10/2018 - 07/2021*

Comcast (via TEKsystems) | Philadelphia, PA

- Delivered four high-visibility initiatives as a lead engineer across Comcast's digital sales and account management platforms, including the Allspark transformation team and Buy Online, Pick Up In-Store across five applications and services.
- Implemented a scalable Consent UI handling millions of legal acceptance records.
- Contributed to the re-architecture of high-traffic customer-facing systems serving millions of users, reducing critical page latency and production defect rates.
- Owned billing, consent, and call-center workflows tied to revenue and compliance, hardening audited workflows against compliance defects.

Managing Partner & CTO *10/2017 - 10/2018*

5 Foot Gorilla, LLC | Philadelphia, PA / Remote

- Co-founded and led a technology consulting firm delivering custom SaaS, web, and mobile solutions from discovery through production launch, owning architecture, execution, and delivery governance.

Principal Software Engineer / Development Manager (Contract) *02/2016 - 09/2017*

Comcast (via FIS Global / Yoh) | Philadelphia, PA

- Helped re-architect Xfinity My Account (about 11 million monthly users) as a lead engineer heading tiger teams. The redesign won the Creativity International Gold Award for Web Redesign.
- Owned billing and payment journeys, among the highest-traffic self-service flows in the application, and partnered across product, design, and service architecture to deliver 26 major releases in 19 months.

Chief Product Officer *05/2015 - 02/2016*

Cnverg | Philadelphia, PA

- Took a prototype-stage product to market in under six months, leading performance tuning and platform hardening that supported 560% user growth and helped secure follow-on venture funding.

User Interface Architect / Tech Lead *11/2013 - 05/2015*

Intuitive Company (now EY Design Studio) | Philadelphia, PA

- Led the technical work on multidisciplinary teams delivering custom enterprise solutions across three concurrent client engagements, planning architecture with principals and engagement leads.
- Delivered reusable UI frameworks that made delivery more consistent, and built client relationships that led to follow-on work.

Owner & Principal Engineer *12/2010 - 11/2013*

Gigabuilder LLC | Philadelphia, PA

- Ran an independent software practice delivering custom SaaS platforms, completing 25+ projects across healthcare, e-commerce, and media with 90% repeat-client business and 95% on-time delivery, bringing on contractors for larger engagements.
- Designed cloud architectures on AWS and Heroku and scaled hosting infrastructure for high-traffic applications.

Senior Front-End Developer *05/2010 - 11/2010*

Band Digital (formerly Whittman-Hart) | Philadelphia, PA

- Built interfaces and maintained hosting for high-profile clients across concurrent projects.

VP of Technology *01/2003 - 01/2005, 01/2006 - 09/2009*

Exam Master Corporation | Philadelphia, PA

- Rewrote Exam Master Online, a high-availability exam delivery platform, and managed IT infrastructure.

User Interface Developer *01/2005 - 01/2006*

cPanel Corporation | Houston, TX

- Redeveloped the cPanel/WHM interface for 12M+ users, improving task completion and time-to-complete by 80%.